Operatig Systems Lab

Lab Task 2

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Step 1: Creating a New User

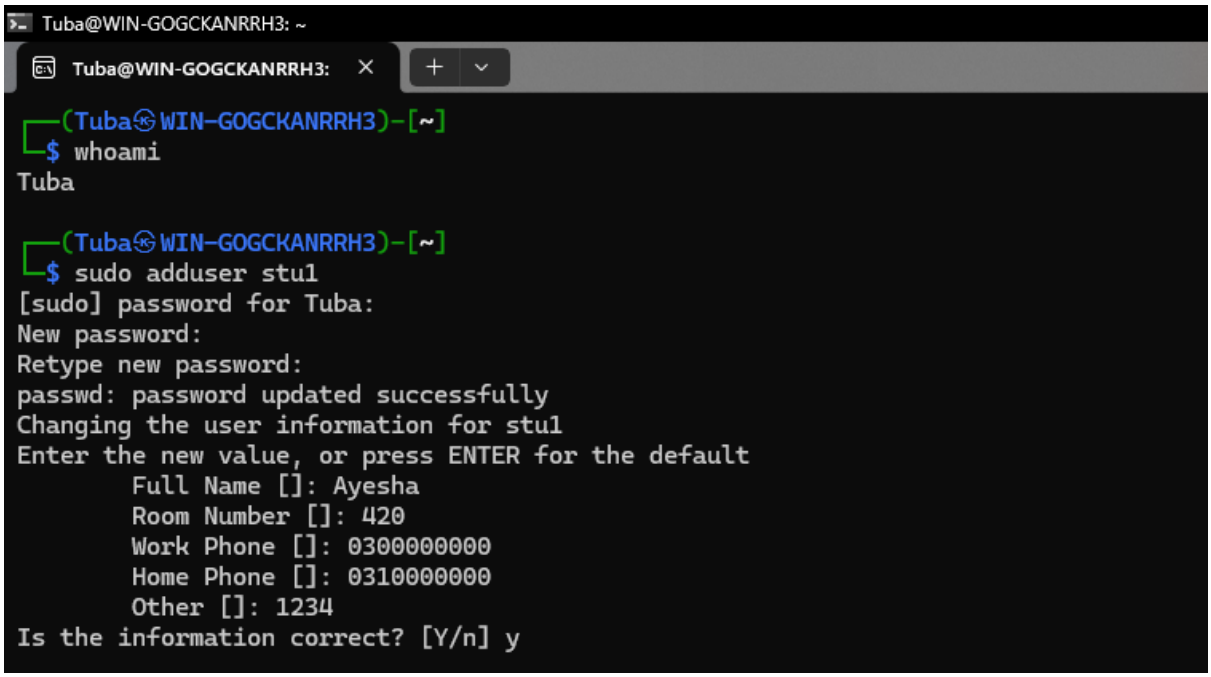
First, I checked my current username by running the whoami command. Then, I created a new user named stu1 using the following command:

sudo adduser stu1

I entered the required password and user details. After creating the user, I verified it using:

id stu1

The output showed the user ID and group information, confirming that the new user was created successfully.



Step 2: Updating an Existing User

In this step, I updated the full name of the existing user stu1 to Ayesha Khan using the usermod command:

sudo usermod -c "Ayesha Khan" stu1

Here, usermod modifies an existing user, and -c updates the comment or full-name field. I then checked the updated information using:

getent passwd stu1

The output displayed Ayesha Khan, confirming that the user's information was updated successfully.

```
┌──(Tuba㉿WIN-GOGCKANRRH3)-[~]
└─$ id stu1
uid=1001(stu1) gid=1001(stu1) groups=1001(stu1),100(users)

┌──(Tuba㉿WIN-GOGCKANRRH3)-[~]
└─$ sudo usermod -c "Ayesha Khan" stu1

┌──(Tuba㉿WIN-GOGCKANRRH3)-[~]
└─$ getent passwd stu1
stu1:x:1001:1001:Ayesha Khan:/home/stu1:/bin/bash
```

Step 3: Opening the Linux Task Manager

First, I updated the package list using the following command:

sudo apt update

Then, I opened the Linux Task Manager using the top command:

top

The top command displayed the currently running processes along with CPU usage, memory usage, and other system information. It helped me monitor the system's performance in real time. I pressed q to exit the Task Manager and return to the terminal.

```
┌──(Tuba㉿WIN-GOGCKANRRH3)-[~]
top - 12:07:50 up 18 min,  2 users,  load average: 0.05, 0.03, 0.00
Tasks: 27 total, 1 running, 26 sleep, 0 d-sleep, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.1 sy,  0.0 ni, 99.9 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   3614.8 total,   3117.0 free,    430.5 used,    140.8 buff/cache
MiB Swap:   1024.0 total,   1024.0 free,      0.0 used.   3184.3 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
     68 root      20   0   39068  13576   9580 S   0.3   0.4   0:00.75 systemd-udevd
      1 root      20   0   26652  16060  12052 S   0.0   0.4   0:00.87 systemd
      2 root      20   0    3804   2512   2376 S   0.0   0.1   0:00.00 init-systemd(ka
      3 root      20   0    3664   2132   2040 S   0.0   0.1   0:00.00 init-watcher
      7 root      20   0    3804   2272   2168 S   0.0   0.1   0:00.00 init
     49 root      20   0   51612  22460  21652 S   0.0   0.6   0:00.23 systemd-journal
    107 root      20   0    4312   2808   2560 S   0.0   0.1   0:00.02 cron
    108 message+  20   0    8324   4728   4068 S   0.0   0.1   0:00.08 dbus-daemon
    110 polkitd   20   0  306800   7972   7144 S   0.0   0.2   0:00.15 polkitd
    115 root      20   0   11348   7496   6640 S   0.0   0.2   0:00.09 systemd-logind
    317 root      20   0    5408   2980   2772 S   0.0   0.1   0:00.01 agetty
    327 root      20   0    7800   4556   4016 S   0.0   0.1   0:00.01 login
    333 Tuba      20   0   18504  12248   9988 S   0.0   0.3   0:00.13 systemd
    335 Tuba      20   0   16164   3488   2156 S   0.0   0.1   0:00.00 (sd-pam)
    361 Tuba      20   0    7424   3792   3544 S   0.0   0.1   0:00.00 mpris-proxy
    365 Tuba      20   0    5336   4900   3360 S   0.0   0.1   0:00.02 bash
    370 Tuba      20   0    8000   4396   4076 S   0.0   0.1   0:00.01 dbus-daemon
    428 root      20   0    3808   1424   1284 S   0.0   0.0   0:00.00 SessionLeader
    429 root      20   0    3876   1492   1284 S   0.0   0.0   0:00.05 Relay(433)
    433 Tuba      20   0    8444   5524   3840 S   0.0   0.1   0:00.10 bash
    486 root      20   0    7804   4540   4000 S   0.0   0.1   0:00.03 login
    496 root      20   0   18244  12276   9996 S   0.0   0.3   0:00.36 systemd
    498 root      20   0   16160   3476   2156 S   0.0   0.1   0:00.00 (sd-pam)
```

Step 4: Killing a Process

First, I started a test process that would run for 300 seconds in the background:

sleep 300 &

The terminal displayed the Process ID (PID) as 855. I then stopped the process using:

kill 855

The terminal displayed a termination message, indicating that the process had stopped. Finally, I verified this using:

ps -p 855

Since no process information appeared for PID 855, it confirmed that the process was no longer running.

